

Search and Rescue AR Visualization Environment (SAVE): Designing an AR Application for Use with Search and Rescue Personnel

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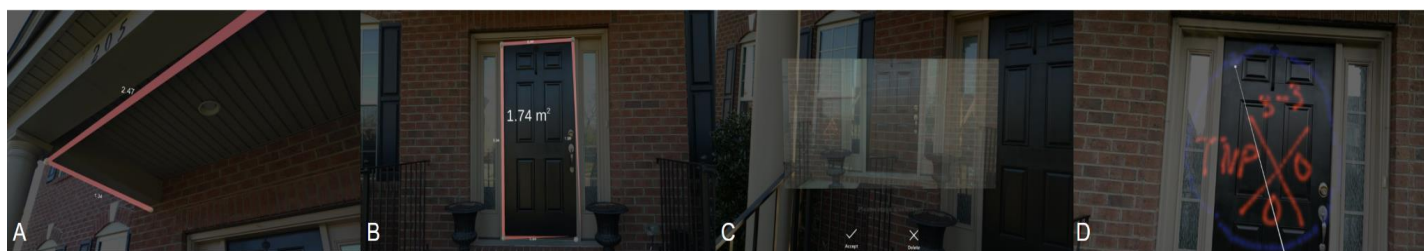


Figure 1: Photo of the (a) line tool (b) area tool (c) picture tool (d) draw tool on real environment

ABSTRACT

Search and Rescue (SAR) operations are often stressful, complex, and time-sensitive operations that require an efficient flow of information between many different teams to operate effectively to save lives. Current augmented reality (AR) technology can be utilized to assist with these operations by helping facilitate information flow while simultaneously making common operation tasks easier to complete. In this paper, we present an early AR prototype, the Search and Rescue AR Visualization Environment (SAVE), which serves as a headworn AR testbed for various SAR features specifically designed for use with ground-based personnel. Throughout the paper, we describe the process of engaging subject matter experts, producing our working testbed prototype, as well as the next steps needed to continue our research.

Keywords: augmented reality, user experience, head-worn display, search and rescue

Index Terms: Human-centered computing — Human computer interaction (HCI) — Interaction paradigms — Mixed / augmented reality;

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1 INTRODUCTION

Today, augmented reality (AR) is growing at an increasingly fast rate with more industrial practitioners and users adopting it every year. This increase in the application of AR is due to its ability to seamlessly integrate additional information, whether that be displaying 3D furniture models in a consumer's living room [1] or layering advanced BIM drawings [2] atop a real construction site, into a user's world with limited disengagement. This theoretically seamless fusion of reality and virtual objects can improve a slew of industries, some of which have seen very small radical technological advancement over the past few decades. One such industry is search and rescue (SAR), which although being tasked with saving people's lives, has not experienced widespread standardized use of AR in the field, especially ground based AR solutions. Due to limited resources, personnel, and extreme conditions, SAR teams generally face difficulties with navigation, communication, search voids and redundancies, acquiring relevant area-related information, and bringing people to safety [3]. Currently, the vast majority of ground SAR personnel rely on walkie talkies, GPS emitters, and cell phones to communicate and plan in urban, wilderness, and wide-area SAR operations, with a few innovative SAR teams using digital maps.

AR lends itself perfectly to SAR, as entire SAR operations' success depends on the gathering, transmission, and understanding of real-time spatial information sent and received by many different users in different locations. More specifically, we see AR's immediate potential for SAR ground personnel, where it can lay dormant if needed, but also display important information when prompted. Even the slightest improvement in the transmission of information using AR could be the difference between life and death in many SAR scenarios, as probability of alive extrication of survivors quickly decreases by the hour [4].

In this paper, we propose the Search and Rescue AR Visualization Environment (SAVE) which aims to support SAR personnel in conducting wide area search operations, which cover extensive urban, suburban and rural land areas, primarily caused

by natural hazards. SAVE, to be utilized by SAR squad members and search facilitators during active deployments, makes use of AR headsets to allow squad members to annotate their environment, set points of interest, and see relevant SAR information. These features will help resolve the difficulties faced in SAR operations, such as conveying the locations of trapped victims or visually defining search areas. In addition, the system will complement squad member's operations by providing more efficient communication between members through AR-supported location markings, photos, and audio to facilitate faster and more productive search responses.

2 RELATED WORK

2.1 Aerial Drone SAR AR

The use of advanced technologies in search and rescue has been, and continues to be, explored to increase the efficiency and success of SAR operations. For example, the incorporation of unmanned aerial vehicles (UAVs), or drones, in SAR provides an aerial perspective on the impacted area. LaLone et al. [5] used drones to fly over and photograph areas of interest, and combined this technology with public mapping platforms for real-time viewing of the images. Drones were utilized to find a spot for the base of operation and to identify areas of interest. Drone contractors painted targets, which were areas the teams on the ground needed to investigate. The targets were visible at the base of operations and in the real world when the SAR team used AR head-worn displays (HWDs). During the operation, each SAR team member wore AR HWDs, which displayed any signal within a mile of them. The signals resembled big pillars of color in the horizon, which thinned to an arrow as they got closer. When members arrived to signal, they labeled it and gave it a priority rating— high priority scenarios would notify responders nearby right away.

Llasag et al. [6] researched the use of drones to aid in human detection in SAR operations for natural disasters. This research explored how to combine interactive interfaces with UAVs and HWDs to provide mixed reality (MR) of AR and VR (Virtual Reality) for the ground operations in SAR. The drones processed information efficiently as computer vision was used to look for humans, and to detect and identify objects. The MR-based interface provided location awareness and video streaming, which resulted in detailed inspection of the area.

2.2 Ground-Based SAR AR

Instances of innovative AR technology can also be seen in ground level aid of SAR operations. One such instance is Wang et al. [7], which utilized an autonomous ground robot, controlled remotely by trained SAR personnel, to search through complex areas in urban SAR scenarios. This autonomous robot used a Microsoft Kinect camera and SLAM software to map the surroundings of the robot and pass the spatial information onto the AR node. This AR node was able to anchor and display virtual markers on mapped surfaces to the remote SAR user. Additionally, the team implemented a neural net computer vision system, using the same Kinect sensor, to automatically highlight and mark designated objects in AR. The designated objects would be trained in the system and over time would become increasingly more accurate in identifying the object in difficult scenarios, including occlusion.

Campos et al. [8] described a collaborative android smartphone app that utilizes AR to assist with on-ground coordination for SAR operations. The solution utilized fixed geolocated AR markers, for locations like command centers and medical locations, overlaid on a user's smartphone camera on the horizon line so they could identify and navigate to them. Markers were synced across all SAR personnel through an off-site server and

allowed for marker assignment to different SAR teams, with availability to set different importance levels for each marker. Additionally, the app showcased a global map view of all current markers and applied an innovative grouping technique to discourage visual clutter at any one time.

3 EARLY RESEARCH

3.1 Intended Hardware

Our team chose to utilize the Microsoft HoloLens 2 as our targeted AR testbed due to its spatial mapping capabilities, ergonomic physical form factor, and its tight integration with the Unity game engine through the Mixed Reality Tool Kit (MRTK) plugin [9]-[11]. Additionally, multi-device HoloLens 2 communication can be coordinated using Microsoft's Azure service, allowing for minimized local computer power when using multiple HoloLens 2 devices in a SAR operation [12]. Finally, Microsoft already sells a hardened hardhat HoloLens 2 device, suited for use in unforgiving environments, like SAR operations [13]. Knowing this intended hardware from the beginning allowed us to :

- Know what capabilities and features we could and could not implement,
- Know device pitfalls that we would need to design around or solve, and,
- Cut down on redundant and exploratory code

All these points together made the subject matter expert interview and final features list portions of the project, to be covered later in this paper, run much smoother and be much more helpful to the advancement of the project.

3.2 Subject Matter Expert Interviews

The subject matter experts (SMEs) interviewed included 7 people, such as Search Team Managers, SAR Instructors, and experienced Search and Rescue Personnel, from Texas, Virginia, and Colorado. We came in contact with these SMEs through the Wide Area Search Training Manager at Texas A&M Engineering Extension Service (TEEX). The interviews provided our team the opportunity to understand SAR in greater detail by learning about technologies currently used, such as GPS and radio, the different roles in an operation, and how hazards are detected and marked. Our team also had the chance to observe some equipment used during SAR operations, such as deep-dive suits and a SAR caravan. The interviews provided insight on shortfalls currently faced during an SAR operation, such as lack of real-time communication, and how information flows in existing SAR systems. These barriers that are often faced, along with others, such as poorly visible map boundaries and lack of detail about information from others, adds unnecessary challenges to operations. This step in our design process allowed us to scope the SAVE AR application to be designed for and used by ground personnel.

3.3 Suggested Features

After completing all SME interviews our team compiled all raw comments from the interviews, in addition to previously conducted background research, and used the three different tools listed below to analyze SAR personnel and their needs [14]:

1. **Work Activity Affinity Diagram (WAAD):** Simplified exercise where the team took all the SME notes and separated them into "buckets" of similar informational affinity over a short period.

2. **Task Structure:** The team organized the SAR tasks and subtasks of a each user work role to granularly see how the tasks fit together.
3. **Flow Model:** The team created a visual representation of the network of people and jobs that are performed on a normal SAR operation and how they fit together to have visual continuity between all user roles.

After we performed these three usage research activities, we were able to determine distinct and separable software features that we could start to implement. The resulting list presented below enumerates these tentative features along with a brief description of how the team envisioned them to work:

- **Line Tool:** Allow users to make lines on surrounding environments and display the relative length of each line,
- **Area Tool:** Allow users to trace a parameter by placing points on environments and displays the area enclosed by the parameter,
- **Picture Tool:** Allow users to take pictures of their environment,
- **Draw Tool:** Allow users to annotate the environment using virtual spray paint,
- **Point of Interest (POI) Tool:** Allows users to mark a particular environment feature with a specific icon as well as insert notes about it,
- **SOS:** A safety button which allows users to indicate when they are in danger,
- **Chat Tool:** Allow users to send direct or group chat messages to the SAR team, and,
- **Map Tool:** A virtual map that shows the position of users as well as search boundaries and other important search information, like POI and markers

3.4 Wireframe Creation

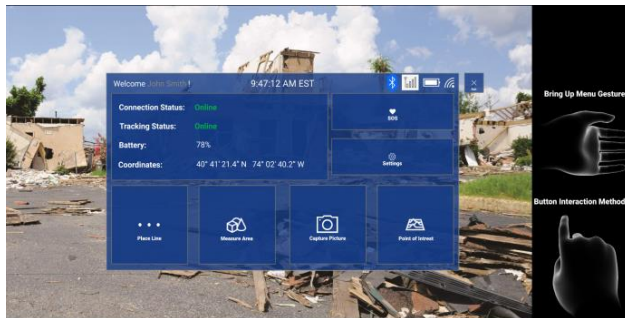


Figure 2: Final interactive wireframe sample

Following our initial SME interviews, our team set out to create interactive AR wireframes for a majority of the features suggested above to test them and the associated user interface (UI) and interactions with SMEs. Figma was used due to its robust collaborative aspect and its accessible cost. Additionally, Figma has an official HoloLens 2 UI kit published by Microsoft which provides premade templates to speed up wireframe design. It was challenging to find the most useful way to create and display a wireframe intended to depict an AR application that interfaces with the physical 3D environment. Further complicating the design process was our decision to utilize hand gesture control for all HoloLens 2 interactions, which would be impossible to integrate interactively in a traditional wireframe software like

Figma. A sample of the completed wireframe is shown in Figure 2.

3.5 Evaluation and Results

The importance of a UX evaluation is that it allows designers to understand how real users may interact with the evolving application user interfaces. Our empirical formative evaluation allowed us to identify usability issues, which can allow for iterations to improve the SAVE application and create a better design to meet user needs. The evaluation entailed a list of key benchmark tasks the SMEs were asked to complete, to identify issues with SAVE interactions. The low-fidelity interactive wireframe that was created became a usable prototype in order to perform the evaluation. This allowed us to get relevant information about gesture interaction and 3D aspects of the system without having to build out a full VR system. The key tasks we asked SMEs to engage in included the use of all system features, such as “Place a line on a damaged building in the environment” and “Send an SOS signal to command base”. Gestures were displayed to show SMEs how to interact with SAVE and a simulated SAR-relevant background was provided for additional context. We asked SMEs to actually perform the gesture as they were clicking through the interactive wireframe. Follow-up questions were presented to the SMEs after the evaluation process such as what frustrations they may have encountered and a feature that is not in the system that they think would be useful. The data collected from these sessions led to a compilation of UX problems to consider during design iterations.

The most unanimous feedback received highlighted the idea of adding urgency values to points of interest. It was also frequently stated that the option of making the system more customizable, such as options for which units of measurements are displayed. A strong piece of feedback to take into consideration was also a tutorial to teach hand gestures so that SAR personnel would be able to use the HoloLens 2 and access features with ease.

4 CURRENT AR TESTBED

This section provides a detailed description of the tools we are currently using to develop SAVE on the HoloLens 2, as well as insight into our current progress and plans for the project. In our quest for creating a unified experience for SAR on the HoloLens 2, we decided to start by laying a strong groundwork of important features. These features, mentioned previously in the “Suggested Features” section, were individually split up and the team has been developing them one by one, in order to flesh out each one.

4.1 Development Tools

The current SAVE system was implemented using Unity 2020.3.16f1 as the game engine and Microsoft Mixed Reality Toolkit v2.7.2.0 Unity plugin to interface with the HoloLens 2 specialty features. All of the coding in Unity was performed in C# using Visual Studio 2019 Community with the SAVE application built as a Universal Windows Platform App (UWP) to be used with the HoloLens 2. Additionally, we utilized the built-in hand gestures provided by the HoloLens 2 and MRTK software to interact with SAVE. Environment mapping was implemented using the HoloLens 2’s spatial mapping described in the next section.

4.2 Device World Tracking



Figure 3: Spatial mesh generated by the HoloLens 2

As previously mentioned, a factor in utilizing the HoloLens 2 for our testbed was the device's ability to easily provide dimensionally accurate and ever-updating environmental data using the built-in spatial mapping feature. The HoloLens 2 uses its suite of visual, time of flight and inertial sensors to provide a spatial mesh of the surrounding environment, as well as the HoloLens 2's relative position with reference to those spatial meshes. The HoloLens 2 is also able to store a number of previously mapped areas locally and can recognize and recall them when needed, which reduces unnecessary mapping of commonly traversed areas. All of this environmental data is passed live to Unity using a "Spatial Awareness" layer, seen in Figure 3, that allows our team to easily build out tools. These tools most commonly use this layer to raycast and spawn holograms on, or with reference to, surrounding manmade and natural environment features proximal to users.

4.3 Current Implemented Features

From the start of our project in August 2021, the team has developed four distinct tools for our SAVE testbed. These include the line tool, area tool, picture tool, and the draw tool. Figure 1 shows examples of each tool, and below we provide an operational explanation of the tool and its purpose.

4.3.1 Line Tool

The line tool (see Figure 1a) allows users to measure straight line 3D distances of the environment around them with ease, similar to the Measure app on iPhone [15]. Users are able to place as many points as they would like and the relative 3D distance between the points will be displayed above the line.

4.3.2 Area Tool

The area tool (see Figure 1b) allows users to place points and get the relative distances between them, similar to the line tool, but additionally displays the approximate 2D area contained within the placed points. This approximate 2D area calculation of a set of possible 3D points is achieved by creating an invisible midpoint, and then creating a triangulated mesh between every point and the midpoint. This resulting mesh is used to approximate area calculations as only simple mesh trigonometry is needed and the area is then displayed to the user in AR in the center of the set of points.

4.3.3 Picture Tool

The picture tool (see Figure 1c) allows users to take pictures of their real surrounding environment and then anchor those photos to the areas in which they were taken. This allows for a fast and easy way to take reference photos in the field and attach them to a physical geographical region where they were taken.

4.3.4 Draw Tool

The draw tool (see Figure 1d) allows users to annotate the environment around them by virtually spray painting the surface using gestures. The spray paint is created on top of the spatial mesh created by the HoloLens 2 so it conforms to environment features and creates a realistic spray painting experience. Additionally, users have the ability to change the size and color of the spray paint and undo their most recent paint drawing feature.

5 DISCUSSION

This project will continue to develop as we finish building out our list of suggested features. The four features presented herein will be further developed, tested and analyzed on our HoloLens 2 testbed to evaluate usefulness and feasibility. We will also continue development on the 4 features we have not built out yet.

Another developing aspect of this project we are looking into is supplementing the spatial mesh and position data of the HoloLens 2 with additional sensors, such as differential GPS. This supplement is being explored due to the HoloLens 2's fairly unreliable large world-scale tracking, especially in featureless, or visually repeating environments, and propensity to lose tracking with quick rotational or translatory movements. We will also explore ways to improve the user experience with SAVE in environments that do not yet have a complete spatial mesh by creating temporary supporting mesh elements by approximation or other methods, in order to limit user frustration when not able to interact with the environment due to lack of generated spatial mesh.

Beyond those additions, we are also working to implement multi-user capabilities to allow virtually placed icons, holograms, and drawings made by different SAR team members be viewable by all connected HoloLens 2 HWDs, providing a truly cohesive team-based solution. This will require additional research into communications systems in remote areas as well. Finally, the entire SAVE system will be tested in real outdoor scenarios with SAR SMEs. This will not only provide us with technological benefits/weaknesses of SAVE, but also show us expert reactions to the solution as a whole.

6 CONCLUSION

In this paper, we have shown our initial thought, design, and prototype process for our SAVE prototype— an AR system implemented on the HoloLens 2 platform aimed to streamline operations for SAR ground personnel. Our early prototype has displayed immense possibilities and potential to solve many of the pain points currently faced by SAR personnel today, in addition to streamlining the SAR operation as a whole. We recognize feedback from our SME-based evaluations, as well as current shortfalls in AR technology and plan to address them in our subsequent prototypes.

Future evaluations include a summative evaluation at TEEX (Texas A&M Engineering Extension Service) contrasting teams with current SAR gear to teams with the SAVE system. Various smaller scale feature specific tests will also be performed in the COGENT Lab. We plan to engage additional SMEs in future research to implement more suggested features, perform more comprehensive testing, and ultimately create a cohesive system we can test in real-world SAR scenarios.

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